

ADVANCED AUTONOMOUS SYSTEMS DESIGN AND OPERATION COMPETITION

2026



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VERSIONS

Table. Versions

VERSIONS	DATE	DESCRIPTION
V 1	02.03.2026	TEKNOFEST 2026 First Version
V 2	13.04.2026	TEKNOFEST 2026 Second Version

DEFINITIONS and ABBREVIATIONS

The abbreviations used within the scope of this specification and their corresponding definitions are provided in this section.

ARB: Advisory and Review Board

UAV: Unmanned Aerial Vehicle

UGV: Unmanned Ground Vehicle

TDZ: Target Detection Zone

UAV-LZ: UAV Launch Zone

UAV-OZ: UAV Operation Zone

UGV-PZ: UGV Positioning Zone

TZ: Target Zone

UAV-LZ: UAV Landing Zone

PRR: Preliminary Review Report

DRR: Detailed Review Report

FRR: Final Review Report

FRS: Final Review Stage

FRP: Final Review Presentation

SRP: System Review Presentation

FRRR: Flight Readiness Review Report

C_{difficulty}: Difficulty Coefficient

w_{UAV}: UAV Weight Score

d_{UAV}: UAV Mission Duration Score

d_{preparation}: Preparation Duration Score

w_{UGV}: UGV Weight Score

w_{payload}: UGV Payload Score

d_{UGV}: UGV Mission Duration Score

S_{mission}: Mission Success

S: Competitor Performance Score

S_{TI}: Technical Inspection Performance Score

S_{flight}: Flight Performance Score

S_{sportmanship}: Sportsmanship Score

D_{deduction}: Deduction Score

1. PURPOSE AND SCOPE

The Advanced Autonomous Systems Design and Operation Competition, organized by Turkish Aerospace Industries (TUSAŞ), aims to guide young talents toward the future of aerospace and space technologies in line with our country's vision of a fully independent defense industry. The competition invites associate, undergraduate, and graduate-level students to become an integral part of our defense industry, while providing them with the opportunity to develop innovative solutions to real-world problems and transform their ideas into prototypes.

It is aimed that the younger generations, who will shape the future of the defense industry, enhance their engineering capabilities, gain experience in complex system design, and acquire the competence to contribute to global advancements in UAV technologies. This initiative is committed to enabling young participants to take an active role in Türkiye's technology initiative and to positioning our country as a center of excellence in this field.

The competitions are organized under the coordination of TUSAŞ within the scope of TEKNOFEST, with the vision of expanding and strengthening the national aerospace ecosystem. The definition of technical rules, supervision of design processes, and performance review are carried out by the Advisory and Review Board (ARB), consisting of expert engineers from TUSAŞ.

This document covers all technical requirements, terms of participation, and the ethical, safety, and technical guidelines to be followed by all participants in the competition.

2. COMPETITION DESCRIPTION

Within the scope of the Advanced Autonomous Systems Design and Operation Competition, teams are expected to autonomously solve a logistics/transportation problem in a simulated post-disaster scenario, through the coordinated operation of an Unmanned Aerial Vehicle (UAV) and an Unmanned Ground Vehicle (UGV). The execution of the mission shall be carried out in accordance with the rules and workflow specified below.

2.1. Competition Area and Operational Environment

The defined zones specified below shall be established within the competition area by the technical teams.

- Target Detection Zone (TDZ): A 50 m × 50 m bounded area within the competition field. This zone contains debris piles and obstacles simulating a disaster environment.
- UAV Launch Zone (UAV-LZ): The area where the UAV catapult system will be installed and from which the UAV will be launched.

- UAV Operation Zone (UAV-OZ): The maximum flight area within the competition field in which the UAV is permitted to operate. This area is limited to dimensions of 300 m × 300 m.
- UGV Positioning Zone (UGV-PZ): The area where the ground vehicle takes its initial position. It is limited to dimensions of 1 m × 1 m.
- Target Zone (TZ): The destination point that the UGV is expected to reach within the target detection zone. It is a 2 m × 2 m area marked with a red “H” symbol.
- UAV Landing Zone (UAV-LZ): The designated safe area where the UAV will land via parachute after completing its mission. It is a 30 m × 30 m blue-colored zone.
- Routes: Within the TDZ, there are three (3) different paths formed between the UGV-PZ and the TZ among the debris piles. On competition days, only one (1) of these three routes will remain open, while the other two will be physically blocked. The information regarding which route is open will not be shared with the participants in advance and will be kept confidential.



Figure 1. Sample illustration of the competition area and operational environment (The image is entirely generated using artificial intelligence. It does not directly represent the actual field.)

2.2. Mission Workflow

For the mission to be considered successful, the system must execute the following steps sequentially and in compliance with the specified rules:

2.2.1. Takeoff

- As specified in the technical requirements, all relevant systems must be transported to the competition area in sealed containers. The setup period shall commence following the opening of these seals under the supervision of the referees.
- The UAV shall be launched only from the designated UAV Launch Zone (UAV-LZ) using a catapult system.

- Hand launch, running takeoff from a runway, or vertical takeoff methods are prohibited; takeoff must be executed exclusively via the catapult system.

2.2.2. Reconnaissance and Route Identification:

- After launch, the UAV shall switch to **fully autonomous** flight mode and operate within the UAV-OZ.
- Using its onboard sensors and image processing algorithms, the UAV shall identify the open route within the TDZ.

2.2.3. Data Transmission:

- The UAV shall transmit the identified route information to the ground station and the UGV via wireless communication systems.
- The field referee at the ground station must confirm in real time that the route information has been successfully detected.
- The field referee at the ground station shall also confirm that the detected route information has been transmitted to the UGV.
- This transmission must be completed while the UAV is airborne.

2.2.4 Landing

- After the route information has been transmitted to the UGV, the UAV shall initiate its landing procedure within the designated UAV-LZ.
- Landing must be performed using a **parachute system**.
- The parachute deployment command may be issued either autonomously by the system or manually triggered from the ground station as a safety precaution.
- If the parachute deployment system is triggered manually instead of autonomously, a deduction score shall be applied to the relevant team.

2.2.5. Ground Operation

- The UGV shall wait for the UAV to complete its landing before initiating movement. The UGV shall not move while the UAV is still in the air.
- Teams whose UGVs begin movement before the UAV has landed shall be deemed unsuccessful in the mission.
- Following the completion of the UAV landing operation, the UGV's deployment process shall be manually triggered by a team member at the Ground Control Station (GCS).
- Based on the route information received from the UAV, the UGV shall autonomously navigate true the open route and proceed toward the TZ.
- Any UGV that does not operate autonomously shall be deemed unsuccessful in the mission.

2.2.6. Mission Completion

- The mission shall be considered successfully completed when the UGV reaches within the boundaries of the TZ by following the route transmitted by the UAV.
- Teams that fail to reach the TZ within the specified time limit shall be deemed unsuccessful in the mission.
- As defined in the technical requirements list, the UGV must transport the designated payload(s) to the target area. Upon arrival at the target point, the UGV must maintain its structural integrity and retain the payload(s) without any loss; otherwise, the mission shall be deemed unsuccessful.

3. COMPETITION SCHEDULE

The schedule prepared for the Advanced Autonomous Systems Design and Operation Competition is presented below.

Table 1. Competition Schedule

Description	Date
Competition Final Application Deadline	10 April 2026
Preliminary Review Report (PRR) Submission Deadline	26.04.2026 – 17.00
Announcement of Preliminary Review Report (PRR) Results	04.05.2026
Detailed Review Report (DRR) Submission Deadline	15.07.2026 – 17.00
Announcement of Detailed Review Report (DRR) Results	29.07.2026
Final Review Report (FRR) Submission Deadline	21.09.2026 – 17.00
Final Review Stage (Final Review Presentations (FRP) and Drone Mission Test Flight)	30.09-03.10.2026
Announcement of Final Review Report Results and Teams Eligible for Financial Support	04.10.2026
System Review Presentation Submission (SRPS) Deadline	05.02.2027 – 17.00
Announcement of SRPS	19.02.2027
Flight Readiness Review Report (FRRR) Submission Deadline	16.04.2027 – 17.00
Announcement of Flight Readiness Review Report (FRRR) Results	30.04.2027
Flight Tests	21.05.2027
Competition Date/Location	August-September 2027

4. COMPETITION PARTICIPATION REQUIREMENTS

- Applications to the Advanced Autonomous Systems Design and Operation Competition are open to all associate, undergraduate, and graduate students who are either Turkish citizens or foreign nationals studying in Türkiye or abroad.
- Teams may consist of students who are either citizens of the Republic of Türkiye or foreign nationals.
- Each team must consist of a minimum of six (6) and a maximum of ten (10) members, including the Team Captain. The number of members permitted to attend the competition site or the TEKNOFEST area may be limited by TUSAŞ and the jury panel. Teams will be informed in advance in case of any such capacity limitations.
- The proportion of graduate-level members within a team shall not exceed 20% of the total team size.
- A participant (student, team captain, or advisor) is not allowed to take part in more than one team within the same competition. In the event of a violation of this rule, all teams associated with the individual in question shall be disqualified from the competition.
- Each team is required to have one (1) Team Captain who holds student status at a university. The active student status of the designated Team Captain must be officially documented.
- The appointment of a mentor/advisor for competing teams is not mandatory and is left to the discretion of the participants. If appointed, each team may include a maximum of one (1) advisor. In the event that this limit is exceeded and more than one advisor is declared, the team's application shall be deemed invalid due to non-compliance with the participation requirements.
- There are no national or international restrictions regarding team formation. Mixed teams consisting of members from multiple countries may either declare their country based on the country represented by the majority of members or choose one of the represented countries at their own discretion.
- As of the application date, all team members except the advisor must have active student status. Teams found to include members who do not meet this requirement shall be disqualified from the competition.
- Team members, excluding the advisor, are required not to be engaged in full-time employment within any institution or organization in the aerospace industry, even if their student status has terminated.
- Originality of the submitted ideas, projects, and systems is essential. Participants may not compete with duplicate or copied reports, ideas, or designs. Any work identified as plagiarism or similarity shall constitute grounds for disqualification.

- Each project may be submitted to only one competition category or one competition track. If the same project is submitted to different categories within TEKNOFEST or to concurrently organized competitions, all applications of the relevant team or individuals shall be deemed invalid.
- The Team Captain is responsible for managing all operational processes, including the application process, preparation of reports, and uploading visual materials to the system, as well as coordinating and representing team members during both the preparation and competition phases.
- Following the Final Review Presentations specified in the competition schedule, no changes or updates to the team's member composition shall be permitted. From this stage onward, team rosters shall be considered final.
- Participants may apply to the competition only after declaring that they have read and accepted all explanations and participation requirements.
- If a participant has previously competed in TEKNOFEST competitions with the same project, they may reapply for TEKNOFEST 2026 provided that the project/idea has been further developed and/or transformed.
- Participants may not compete with an exact copy or duplicate of a previously submitted project/report/idea. Projects identified as similar or plagiarized shall be disqualified.
- In cases where project reports from previous TEKNOFEST competitions published at www.teknofest.org are referenced, proper citation must be provided. The citation format can be found under the general rules section of the specification.
- Participants may apply to different TEKNOFEST competitions with different projects.
- When selecting a category, the participant's current level of education at the time of application must be taken into account.
- If the participant has previously taken part in another competition (TEKNOFEST or otherwise) with the same project, they must declare the name, location, date, organizer, and result of that competition within the project documentation.
- Submitted ideas and documents are expected to be original and free from plagiarism. All submitted reports will be screened using plagiarism detection tools during the competition process.
- Consistent use of verb tenses (e.g., present tense, past tense, etc.) is required in application texts.
- Reports must not contain irrelevant, manipulative, or political statements unrelated to the subject of the project.
- In case of non-compliance with the rules or procedural violations, the team/participant shall be directly disqualified.

- Disqualified participants shall not have the right to appeal competition results or committee decisions.
- The TEKNOFEST Competitions Committee reserves the right to issue additional announcements or update the rules during the competition period.
- All participants are responsible for following announcements.
- Students are required to upload officially approved student certificates, and advisors must upload official documents verifying their place of employment, bearing wet signatures, to the system on the date to be announced by the TEKNOFEST Competitions Committee. (Document templates will be shared with participants.)
- Applications must be submitted online via the www.t3kys.com application system by April 10, 2026.

4.1. Team Formation

- Teams must consist of a minimum of 6 and a maximum of 10 members across all categories and education levels. (The advisor is not included in this count.)
- There is no mandatory requirement for teams to have an advisor.
- The advisor must not be registered as a team member. The advisor's role within the team must be explicitly indicated as "advisor."
- Each team may have a maximum of one advisor. Applications from teams with more than one advisor shall be deemed invalid.
- Teams may be formed by students from a single higher education institution or by students from multiple higher education institutions.
- Teams may be formed by students from a single country or by students from one or more countries. The team's country information may be selected based on the country with the majority of its members, or the team may choose any of the represented countries at its own discretion.
- Each team must have a Team Captain.
- Throughout the competition process, the education level selected during the application phase will be taken into consideration when forming the team. Participants must ensure that their selected education level is accurate.
- During the application period, the Team Captain and/or advisor (if any) registers on the system, creates the team profile with complete and accurate information for all team members, and sends invitations to the members' email addresses. Invited members must log into the application system, accept the invitation under the "My Team Information" section, and complete their registration. Otherwise, the registration process shall not be considered complete.

4.2. Advisor Responsibilities

- The document confirming that the person assigned as an advisor will fulfill their advisory duties must be uploaded to the system with a wet signature on the date to be announced by the TEKNOFEST Competitions Committee
- Participants are not required to attend the final stage with their advisors. However, if a team advisor is included in the application, submission of the advisory commitment document is mandatory.
- The role of the advisor is to assist students in planning their own education, to guide them on academic, social, and cultural matters, and to help create an environment conducive to the holistic development of the student's intellectual, social, and emotional aspects. The advisor's role within the team is to provide academic support as needed and to guide team members in developing solutions to problems encountered in the project.
- Advisor eligibility: Individuals who are employed as teachers or academics in educational institutions, or professionals such as engineers/experts actively pursuing their careers in relevant fields, may serve as advisors within the team.
- The advisor commits to supporting the team until the final stage and to being present with the team throughout the final stage (except flight area/airfield) of the competition.

4.3. Process Information

- Throughout the competition process, all communications issued by the TEKNOFEST Competitions Committee shall be directed to the person designated as the team's communication officer. Therefore, each team must appoint a communication officer. (Notifications are sent to the email address registered in the KYS system.)
- The tracking of all processes (application submission, report/presentation/form final upload deadlines, objection processes, required forms, etc.) is the responsibility of the communication officer. The TEKNOFEST Competitions Committee shall not be held responsible for any delays or issues arising from the communication officer.
- All necessary processes within the scope of the competition (application, report/presentation/form submissions, review results, financial support applications, objection procedures, member additions/removals, etc.) are conducted via the KYS system. Teams are required to monitor all processes through the KYS portal.
- During the competition process, application submission, report uploading, objection procedures, and form completion shall be carried out under the authority of the Team Captain and/or advisor, and all competition processes are managed through these individuals.
- Member additions and removals may be carried out until the Final Review Report submission deadline. No changes to team composition are allowed after report submission. The TEKNOFEST Competitions Committee may permit changes to team

members up to a specified date; after this date, no changes to the team structure shall be allowed.

- Travel and accommodation support provided to finalist teams is limited. The TEKNOFEST Competitions Committee reserves the right to change the number of supported participants. This limitation must be considered during team formation.
- The TEKNOFEST Competitions Committee reserves the right to limit the number of team members allowed at the festival area. In case of such limitation, teams will be informed by the committee.
- Participants who have completed team formation must apply to the appropriate competition category relevant to their project.
- Participants who complete the application process are required to prepare and upload their forms in full to the system by the Preliminary Design Report final submission deadline specified in the competition schedule, without being subject to any elimination process. Teams that fail to upload their reports within the specified timeframe shall be deemed eliminated at this stage.

4.4. Application Principles

Applications shall be submitted via the official TEKNOFEST Aviation, Space and Technology Festival Technology Competitions website www.teknofest.org.

Applications must be completed online through the www.t3kys.com application system by April 10, 2026.

Participants may apply to the competition only after reading and accepting all statements and participation requirements related to the competition.

Applicants shall be deemed to have accepted all conditions set forth in this specification upon submission of their application.

5. COMPETITION TECHNICAL PROJECT PROCESS

5.1. Preliminary Review Report (PRR)

The Preliminary Review Report (PRR) to be prepared for the Advanced Autonomous Systems and Operation Competition constitutes the first critical stage of the competition's technical project process. The report consists of a total of eight (8) main sections, namely Summary, Organizational Structure, Project Schedule, Budget Analysis, Conceptual Design Approach, Benefit Analysis Studies, Production Plan, and System Integration and Test Plan. The sections and main headings included within the scope of the PRR are presented in Table 1.

Teams are required to present all preparation and conceptual processes in an analytical manner and in alignment with the success criteria specified in Table 1. This is a mandatory prerequisite for advancing to the next technical phase. The guideline document prepared for the Preliminary Review Report will be shared with participants who are eligible to proceed to the competition technical project process following the completion of the initial application intake and review process.

Table 2. Preliminary Review Report (PRR) Assessment Criteria

Chapter No	Subject	Score
Chapter 1	Summary	10
Chapter 2	Organizational Structure	5
Chapter 3	Project Schedule	5
Chapter 4	Budget Analysis	5
Chapter 5	Conceptual Design Approach	30
Chapter 6	Benefit Analysis Studies	25
Chapter 7	Production Plan	10
Chapter 8	System Integration and Test Plan	10

5.2. Detailed Review Report (DRR)

The second technical phase of the Advanced Autonomous Systems and Operations Competition is the Detailed Review Report (DRR). The DRR consists of five (5) main sections: Summary, Organizational Structure and Project Schedule, General System Specifications, Test Activities, and Requirements Traceability and Review. This report must provide detailed information regarding the UAV conceptual design and the actions taken within the scope of structural and hardware/software activities for the UAV-UGV-Catapult systems. Any iterations conducted within this scope must be presented in detail. In addition, participants are expected to provide an analytical approach regarding the traceability and review of competition requirements.

Table 3. Detailed Review Report (DRR) Assessment Criteria

Chapter No	Subject	Score
Chapter 1	Summary	10
Chapter 2	Organizational Structure and Project Schedule	10
Chapter 3	General System Specifications	60
Chapter 4	Test Activities	10
Chapter 5	Requirements Traceability and Review	10

Teams deemed successful according to the DRR results will be entitled to receive 25% of the total budget if they have applied for financial support. In this context, it is necessary to complete the financial support form, which will be submitted via the KYS system.

The date for submitting the financial support form will be determined by the TEKNOFEST Competitions Committee, and the relevant team's communication representative will be notified.

Note: Applying for financial support is not mandatory.

5.3. Final Review Stage (FRS):

This stage of the competition is a critical qualifying step where teams partially validate their theoretical designs and software architectures under field conditions. This stage consists of two sub-sections:

5.3.1. Final Review Report (FRR)

Final Review Report (FRR), is a technical document that shall be prepared to demonstrate that all detailed design processes of the relevant systems have been completed and that the system is ready to meet the specified mission requirements. Within the scope of the FRR, evidence shall be provided to verify that the requirements defined during the Preliminary Review Report (PRR) and Detailed Review Report (DRR) phases have been satisfied, that all necessary revisions have been implemented, that the design is manufacturable, and that system integration has been completed. All revisions made throughout the process shall be documented and presented within the FRR. Furthermore, in order to demonstrate that the system is capable of operating under actual mission conditions, the final mechanical, avionics, and software architectures, subsystem interactions, as well as manufacturing, assembly, and field deployment processes shall be described in detail. The sections, main headings, and review criteria defined within the scope of the FRR are provided in Table 3.

Tablo 4. Final Review Report (FRR) Assessment Criteria

Chapter No	Subject	Score
Chapter 1	Summary	5
Chapter 2	Organizational Structure and Project Schedule	5
Chapter 3	General System Specifications	40
Chapter 4	Test Activities	15
Chapter 5	Manufacturing Activities	15
Chapter 6	Requirements Traceability and Review	10
Chapter 7	Budget Performance Analysis	10

5.3.2. Drone Mission Test Flight

This phase is a mandatory flight phase in which the mission software and autonomy algorithms developed by the teams for the competition are tested under field conditions. Teams are required to perform this mission using a multirotor unmanned aerial vehicle (Drone) capable of vertical take-off and landing, integrated with their unique hardware and software architectures. The mission shall be conducted within a simulated 20×20 meter test area containing obstacles and debris structures. Within the scope of the mission, the drone is required to detect the “Open Path” that provides a safe transition between the designated “Start Zone” and “Target Zone” by flying at a minimum altitude of 30 meters. This detection shall be performed using aerial image processing and artificial intelligence algorithms, and the identified path shall be transmitted in real time to the Ground Control Station (GCS) during flight. In order for the mission test flight to be considered successful and for teams to receive the full score of 100 points, the real-time path detection must be verified and approved by the field referee, and the entire operation process (take-off, detection, data transmission, and landing) must be completed within a maximum of 10 minutes. Teams that fail to meet any of the specified criteria, exceed the time limit, or perform incorrect path detection shall receive a score of 0 and will be considered unsuccessful in the mission. The total score for the final review phase is calculated according to the following weighted average formula:

$$\text{Final Review Stage Total Score} = (\text{FRR} * 0.85) + (\text{Drone Mission Test Flight} * 0.15)$$

Based on the results of the Final Review Stage (FRS), teams that are deemed successful shall be entitled to receive 75% of the total budget, provided that they have submitted a financial support application. Within this scope, the financial support form to be submitted through the KYS system must be completed in full.

The deadline for the financial support form shall be determined by the TEKNOFEST Competitions Committee, and the relevant information shall be communicated to the designated contact person of the respective team.

Note: *Submitting a financial support application is not mandatory.*

5.4. System Review Presentation (SRP):

System Review Presentation is a mandatory presentation document to be prepared by teams for the purpose of verifiably validating the design/manufacturing/procurement maturity of the UAV, UGV, and Catapult systems developed within the scope of the competition. The BOM (Bill of Materials); which is the basis of the presentation, is the official materials list that includes all manufactured or purchased parts, subsystems, and consumables of the system (e.g., adhesives/binders), listing them with their names, quantities, and basic information. The presentation shall include, for each part and/or commercially procured system item included in the BOM, the manufacturing method,

manufacturing duration, CAD model mass, actual measured mass after manufacturing or delivery, any additional fixture/tool requirements for manufacturing, and any manufacturing difficulties encountered; for commercially procured subsystems, procurement status (received / ordered / no action taken yet) along with the mass and dimensional information of the purchased items shall also be reported. In this way, the entirety of the declared BOM shall be checked on a part/system basis, and progress tracking shall be standardized. No additional scoring shall be conducted after the presentation, and the evaluation shall be announced as “Successful / Unsuccessful.”

5.5. Flight Readiness Review Report (FRRR):

Within the scope of this report, competitors shall present in detail the final images, physical characteristics (such as dimensions, weight, etc.), and total budget analyses of the assembled UAV and UGV systems. Differences between the actual project schedule and the project schedule projected in the FRR shall be provided with detailed explanations. Any changes applied to the assembly and integration steps planned in the FRR shall be explained along with their justifications.

Competitors shall also include, within the report, videos demonstrating the successful autonomous operations of the UAV, UGV, and catapult systems prior to the test flight. The algorithms and models to be used for mission execution shall be described in detail in their final forms within the report.

Table 5. *Flight Readiness Review Report (FRRR) Assessment Criteria*

Chapter No	Subject	Score
Chapter 1	Summary	5
Chapter 2	Project Schedule	5
Chapter 3	Budget Performance Analysis and Deviation	10
Chapter 4	General System Specifications	40
Chapter 5	Flight and Autonomous Operation Videos	40

6. COMPETITION IMPLEMENTATION PRINCIPLES

- Within the scope of the Advanced Autonomous Systems Design and Operation Competition, it is mandatory that the aerial vehicle platforms to be developed are configured as fixed-wing unmanned aerial vehicles (UAVs).
- The term “system” used throughout all guideline documents collectively refers to Unmanned Aerial Vehicles (UAVs), Unmanned Ground Vehicles (UGVs), and launch (catapult) platforms in an integrated manner.
- It is essential that all design, sizing, and manufacturing processes of the systems submitted to the competition are carried out entirely by the competing teams. Teams

participating with commercially available off-the-shelf platforms or designs found to contain plagiarism/copying shall be disqualified from the competition.

- The maximum weight limit defined for aerial vehicles to be produced within the scope of the Advanced Autonomous Systems Competition is 25 kg, including operational payloads. Vehicles exceeding this limit shall be excluded from evaluation during the technical inspection phase.
- The maximum weight limit defined for ground vehicles to be produced within the scope of the Advanced Autonomous Systems Competition is 10 kg, excluding operational payloads. Vehicles exceeding this limit shall be excluded from evaluation during the technical inspection phase.
- The dimensions of the operational flight area and the field configuration shall be announced to all teams through official channels after the competition venue is determined.
- The relevant institution reserves the right to make changes, updates, or revisions to this specification and all related technical documents throughout the competition process in accordance with operational requirements, technical needs, or force majeure conditions. Any such changes shall become binding for all participating teams from the moment they are announced via the official website and communication channels of the competition.
- The competition shall last a total of 6 (six) days. This 6-day period shall be divided into 3 (three) separate task periods, each consisting of two (2) consecutive days.
- Within the target detection area, there are 3 (three) different routes that provide passage between the UGV Positioning Zone (UGV-PZ) and the Target Zone (TZ). These routes are designed in three different difficulty categories: “Easy,” “Medium,” and “Hard.” Each route shall have a different difficulty coefficient in evaluation calculations.
- In each two-day period of the competition, only one (1) of these three routes shall be available for use, while the other two routes shall be completely closed with physical obstacles (barriers, debris simulations, etc.). Information regarding which route and difficulty level will be open on which competition days shall not be shared with competitors in advance and shall remain confidential.
- Due to the uncertainty of the open route, teams are required to use sensors and image processing technologies during the UAV’s autonomous flight to scan physical obstacles in the field and determine the accessible route for the UGV to proceed.
- Each competitor team shall have only one (1) flight attempt for each route made available (each 2-day period). Accordingly, teams shall have a total of three (3) flight attempts over the 6-day competition period.
- The flight attempts available to teams shall only be valid within the period of the currently open route. Any unused flight attempt within the designated period of a given route shall not be carried over to subsequent routes or other days under any

circumstances. Unused flight attempts shall be forfeited, and the team shall receive a score of zero (0) for the mission of that round.

6.1. Before the Competition

- Competitor teams are required to complete their on-site registration procedures on the first day of the event week. The Team Captain must be personally present during registration, and the registration procedures shall be carried out by the Team Captain at the Management Office. In the event that team members who are not present during the registration hours arrive at the field later, the Team Captain, together with the respective member, is obligated to apply to the Management Office to report the situation and update the final registration form.
- Teams that have completed on-site registration shall apply to the Technical Inspection Tent to obtain technical compliance approval.
- All teams that have completed the registration process are required to undergo the technical inspection process prior to proceeding to the flight phase. No team that has not completed the technical inspection phase shall be permitted to conduct flight operations.
- All decisions made by TUSAŞ officials and the jury panel during the technical inspection phase are final and binding. Participating teams are required to comply with these decisions unconditionally throughout the inspection process.
- It is essential that teams undergo technical inspection with the system configurations they have declared and committed to within the scope of the Final Evaluation Presentation, System Review Presentation, and Flight Readiness Assessment Report. Deviations from the reported technical specifications or participation with a different system shall be considered a violation of the rules.
- The storage cases of UAV and UGV systems that successfully pass technical inspection and receive “Flight Approved” status shall be sealed by the referees. Once sealed, any modification, disassembly, or part replacement in the mechanical or electronic structure of the systems prior to the mission is strictly prohibited for safety reasons.
- Teams whose storage cases are sealed following technical inspection shall not interfere with these seals until their assigned flight sequence. The opening of the seals shall be carried out at the flight area and immediately prior to the start of the preparation period, under the supervision of the jury panel.

6.2. Competition Sequence

- Competitive teams have the right to determine their flight order based on the ranking resulting from the Final Evaluation Presentations. The team that achieves the highest score in the evaluation will have priority in selecting its position in the flight schedule.

- No objections regarding grouping or flight order will be accepted. The established schedule is final, binding for all teams, and cannot be altered in order to ensure uninterrupted competition operations.
- All equipment, tools, and materials required for the execution of tasks during the competition are the responsibility of the teams.
- The responsibility for ensuring that all systems and supporting infrastructure components to be used in operational processes are available on-site lies with the teams.
- In all possible risk scenarios, the jury has the authority to terminate the operation in order to maintain flight safety. In the event that the UAV sustains any damage or failure prior to or during flight, the system must undergo a comprehensive technical inspection. Platforms that do not receive technical compliance approval as a result of this inspection will not be permitted to continue flight operations under any circumstances.
- No appeal mechanism shall be operated for flight operations that are rejected or terminated on safety grounds. Administrative and technical decisions taken with a safety-oriented approach are binding for all participants; any applications made in this regard shall be deemed invalid.
- A maximum of eight (8) team members from each team are allowed to enter the competition area.
- Advisors (Academic Advisors or Mentor Teachers) are strictly prohibited from entering the flight area.
- After entering the competition area, teams are not permitted to communicate with their advisors outside the area by verbal, telephone, or any form of signaling. If detected, the team will be disqualified.
- All team members present in the area during flight are required to wear the provided helmets and vests.
- Teams enter the competition area upon the approval of the “Area Referee.” With this approval, the team’s 15-minute “Preparation Time” begins.
- During the preparation time, teams must complete the following: UAV assembly, placement of the catapult system inside the UAV launch zone (UAV-LZ), mounting of the UAV onto the catapult, placement of the unmanned ground vehicle (UGV) into the UGV positioning zone (UGV-PZ), and setup of the Ground Control Station.
- The team captain declares “Team Ready” to the referee upon completion of preparations, at which point the preparation time is stopped.
- After the “Team Ready” declaration, any physical intervention on the vehicles or the ground station by participants is prohibited.
- In case intervention on the systems is required due to a technical malfunction, the team captain may proceed by declaring to the referee that they accept a 100-point

penalty. The time spent for intervention is added to the team's preparation time. If the total allotted time is exceeded, the flight right is revoked.

- After preparations are completed, “Flight Readiness” approval must be obtained from the Area Referees. No launch may be conducted without this approval.
- The flight time begins when the UAV separates from the catapult system (launch) and ends when route determination, data transmission, and parachute landing are completed. The maximum time allowed for these operations is 15 minutes.
- After launch via the catapult system, if the UAV fails to remain airborne and contacts the ground; provided that the Area Referee confirms that no incident or structural damage compromising flight safety has occurred, the team is granted a re-launch right.
- The time spent for the re-launch process (retrieving the vehicle, re-installing it on the catapult, and launching again) is not paused and is included within the total flight time (15 minutes). Teams must complete this process within their remaining task time. No launch may be performed once the time has expired.
- During re-setup, only catapult installation and system restart procedures are permitted. Comprehensive repairs or component replacement on the vehicle are strictly prohibited during this process.
- The open route detected by the UAV must be transmitted in real time to the Ground Control Station and monitored by the Area Referee via the screen. The referee must verbally confirm that the detection has been successful. Any detections not approved by the referee shall be deemed invalid.
- The UAV must conduct its flight within the defined UAV operational boundaries (UAV-OB). Flights exceeding the UAV-OB limits shall be considered a “Boundary Violation,” and the flight will be deemed unsuccessful.
- The unmanned ground vehicle (UGV) may move toward the Target Zone (TZ) only after a successful UAV mission (i.e., transmission of route data and completion of the UAV's landing), based on the data received from the UAV. The maximum time allowed for the UGV to reach the TZ is 10 minutes. If this time is exceeded, the UGV task will be considered unsuccessful.
- Both the UAV and the UGV must perform all processes—from launch/start to mission completion (including navigation, detection, data transmission, and driving)—fully autonomously.

6.3. After the Competition

- Upon completion of flight operations, all teams are obliged to promptly remove their systems and equipment from the flight area. Participants are responsible for the swift collection of equipment to ensure the field is prepared for the next flight.

- Any verbal or physical interference directed at TUSAŞ and TEKNOFEST officials, the jury panel, or support personnel on the competition field is strictly prohibited. If such disciplinary misconduct is detected, the team associated with the individual(s) involved may be subject to disqualification from the competition.

6.4. Conditions for Flight to be Deemed Invalid

Mission flights performed during the competition shall be deemed “Invalid,” and no points shall be awarded for the relevant flight round if any of the following conditions occur:

- **Time Exceedance:** The UAV mission time begins when the UAV separates from the catapult system and ends upon completion of route determination, data transmission, and parachute landing. The maximum time allowed for these processes is 15 minutes. If this duration is exceeded, the flight shall be deemed invalid even if the mission is completed.
- **Manual Intervention:** If, at any stage of the flight (takeoff, cruise, or landing), the Safety Pilot intervenes and switches the UAV to manual mode due to safety concerns or a technical malfunction, the flight shall be considered unsuccessful.
- **Referee Verification and Data Transmission:** The route detected by the UAV must be transmitted in real time to the Ground Control Station. If the transmitted data cannot be verified by the Area Referee, or if data transmission is not maintained, the flight shall be deemed invalid.
- **Boundary Violation:** If the UAV exits the UAV operational boundaries (UAV-OB) defined by the competition committee, the flight shall be deemed unsuccessful in accordance with safety procedures.
- **Crash and Structural Integrity:** If the UAV sustains a crash, breaks apart, or loses its structural integrity at any stage of the flight or during landing (except for propeller breakage), the flight shall be deemed unsuccessful.
- **Landing Violation:** If, during the UAV’s landing within the designated UAV Landing Zone (UAV-LZ) after mission completion, the parachute system fails to deploy, resulting in a hard landing (uncontrolled descent), the flight shall be deemed invalid.

7. REVIEW and SCORING

All review and scoring activities to be conducted throughout the competition schedule shall be carried out based on the formulations presented below. Within this scope, a technical competency-based, multi-stage elimination mechanism has been established, encompassing the Preliminary Review Report (PRR), Detailed Review Report (DRR), and Final Review Stage (FRS) phases in their entirety.

$$Score_1 = 0.1 \times PRR + 0.25 \times DRR + 0.55 \times FRS + 0.1 \times FRRR$$

$$Score_2 = C_{\text{difficulty}}$$

$$\times \left[S_{\text{mission}} + \frac{25000 - w_{UAV}}{10} + \frac{900 - d_{UAV}}{5} + \frac{900 - d_{\text{preparation}}}{30} + \frac{10000 - w_{UGV}}{15} + (w_{\text{payload}} \times 0.01) + \frac{600 - d_{UGV}}{15} - D_{\text{deduction}} \right]$$

$$C_{\text{difficulty}} = \begin{cases} 1.0 & \text{Scenario 1} \\ 1.5 & \text{Scenario 2} \\ 2.0 & \text{Scenario 3} \end{cases}$$

$$S = S_{TI} + S_{\text{flight}} + S_{\text{sportmanship}}$$

$$Score = 0.65 \times Score_2 + 0.20 \times S + 0.15 \times Score_1$$

Table 6. Competition Review and Scoring Criteria

Symbol	Parameter Name	Calculation Method / Description
Score₁	Report Performance Score	This is the score obtained by the competitors based on the reports they prepare and the presentations they deliver throughout the entire process.
PRR	Preliminary Review Report Score	This is the score obtained by the team as a result of the evaluation of the Preliminary Review Report.
DRR	Detail Review Report Score	This is the score obtained by the team as a result of the evaluation of the Detailed Review Report.
FRS	Final Review Stage Score	This is the score obtained by the team as a result of the evaluation of the Final Review Stage.
Score₂	Flight Performance Score	This is the score obtained by the competitors based on their mission flight performance.
C_{difficulty}	Difficulty Coefficient	This is the coefficient that varies depending on the scenario in which the mission is carried out.
w_{UAV}	UAV Weight Score	1 point is awarded for every 10 grams below the 25,000-gram limit of the UAV.
d_{UAV}	UAV Mission Duration Score	1 point is awarded for every 5 seconds saved from the UAV's maximum mission duration of 900 seconds. The UAV's mission time begins with its launch from the catapult and ends when it completes its parachute landing.
d_{preparation}	Preparation Duration Score	1 point is awarded for every 30 seconds saved from the maximum setup time of 900 seconds. The setup time begins with the signal of the assigned referee and ends when the team

		signals that they are ready (Hands-Off). After notifying the referee that they are ready, if teams need to intervene in their systems again, they may do so by accepting a penalty point, provided that the duration of such intervention is added to their preparation time.
w_{UGV}	UGV Weight Score	1 point is awarded for every 15 grams below the 10,000-gram limit of the UGV.
$w_{payload}$	UGV Payload Score	1 point is awarded for every 100 grams of total payload carried by the UGV.
d_{UGV}	UGV Mission Duration Score	This is the total mission duration of the UGV. The maximum mission time of the UGV is 600 seconds. The time begins when the UGV's wheels start moving and ends when the UGV physically contacts the tarp. The UGV is awarded 1 point for every 15 seconds saved below the 600-second limit.
$s_{mission}$	Mission Success	This is the score awarded if the UGV physically contacts the target tarp (Point H). In case of success, 600 points are awarded; if unsuccessful, 0 points are awarded.
s	Competitor Performance Score	-
s_{TI}	Technical Inspection Performance Score	This is the score obtained if the team successfully completes all technical inspections. If the team passes the technical inspection on the first attempt, 100 points are awarded; if on the second attempt, 50 points are awarded; and if the team fails to pass the technical inspection, 0 points are awarded. Teams that do not pass the technical inspection are not allowed to perform a flight.
s_{flight}	Flight Performance Score	This is the score obtained if the team performs a successful flight in accordance with the rules. For a successful flight, the UAV must not be damaged, must be able to detect the target, and must enable the UGV to move. The team is awarded 200 points for each scenario flight completed successfully; otherwise, 0 points are awarded.
$s_{sportmanship}$	Sportmanship Score	This is the score obtained based on the team's teamwork and sportsmanship demonstrated throughout the competition. A maximum of 50 points can be awarded.

FRRR	Flight Readiness Review Report	This is the score obtained by the team as a result of the review of the Flight Readiness Review Report.
D_{Deduction}	Deduction Score	Landing Deviation: If the UAV lands outside the designated landing area, a penalty of -5 points will be applied for each meter of deviation from the area. Restart: If teams need to intervene in their systems after the hands-off stage during the preparation phase, a penalty of -100 points will be applied. Manual Parachute Triggering: If the UAV's parachute system is activated manually during landing, a penalty of -50 points will be applied

8. AWARDS

The details of the awards to be presented to the teams ranking in the top three at the end of the competition are specified in Table 7.

Table 7. Competition Awards

Ranking	Prize Amount	Advisor
1st Place	750,000 TL	50,000 TL
2nd Place	600,000 TL	40,000 TL
3rd Place	500,000 TL	30,000 TL

8.1. TUSAŞ Special Award

- Members of the teams that rank in the top three (3) in the overall competition standings will be directly granted the right to participate in the summer internship program to be held at the TUSAŞ Ankara campus. This right shall be valid provided that the relevant students meet TUSAŞ's existing internship eligibility criteria.
- In addition to the internship opportunity, if any additional awards are defined, scope and schedule of such awards will be announced to the participants at a later stage through official channels.

9. ADMINISTRATIVE PROVISIONS

- Throughout the competition schedule, all participating teams are obliged to continuously follow announcements published on the official website teknofest.org and notifications sent to the email addresses registered in the

application system. All information communicated through these channels shall be considered as official notification.

- TUSAŞ reserves the right to modify the provisions of the specifications and update clauses whenever deemed necessary. The most recent version of the specifications announced through official channels during the competition process shall prevail; previous versions shall be considered invalid.
- Situations such as insufficient or delayed funding, disruptions in customs procedures, and delays in logistics/cargo shipments—being foreseeable and preventable with prior precautions—shall not be accepted as valid excuses (force majeure) for failure to participate in the competition.
- It is essential that there is no first-degree familial relationship, direct benefit relationship, or any conflict of interest between the students applying to the competition and, if applicable, the team advisor. In accordance with ethical principles, if such situations that may compromise impartiality are detected, the relevant applications shall be rejected without evaluation.
- Any unethical, discourteous, or aggressive behavior exhibited by any member of the participating teams—towards TUSAŞ personnel, TEKNOFEST officials, members of the Advisory and Review Board (ARB), the jury panel, other teams, or field staff—whether in writing, verbally, physically, face-to-face, or via digital platforms at any stage of the competition (before, during, or after) is strictly prohibited. If such behavior is detected, the relevant team may be disqualified from the competition by TUSAŞ and TEKNOFEST and may be banned from all competitions for a period to be determined.

10. CONTACT

For questions regarding the competition, you may join the competition group from the “Advanced Autonomous Systems Design and Operation Competition” page on the TEKNOFEST website. It is the responsibility of the teams to actively follow this group, and at least one member from each team must join and track announcements, questions, and answers shared in the group. The TEKNOFEST Competitions Committee shall not be held responsible for any failure of teams to access up-to-date information due to not following the specified email/group communications.

Questions related to the organizational aspects of the competition should be submitted via the email address iletisim@teknofest.org

Submitting your questions through the correct channels mentioned above is important to ensure that they are addressed promptly.

11. GENERAL RULES

To access the General Rules booklet applicable to the competition, please [click here](#).

12. ETHICAL RULES

To access the Ethical Rules booklet applicable to the competition, please [click here](#).

Statement of Responsibility

T3 Foundation and TEKNOFEST shall not be held responsible in any way for any injury or damage arising from any product submitted by the competitors or caused by the competitors themselves. T3 Foundation and the organizers are not responsible for any damages caused by the competitors to third parties. T3 Foundation and TEKNOFEST are not responsible for ensuring that teams develop and operate their systems in accordance with the laws of the Republic of Türkiye.

Türkiye Technology Team Foundation reserves the right to make any changes to this specification.

